

ECON 21030: Honors Econometrics

Panel Data

Spring Quarter 2024

Crossing your *is*, dotting your *ts*

So far, we have been thinking about data generating processes where each random variable, e.g. Y_i , captures an attribute of a single unit (the i). We have discussed examples where Y_i might represent the income of individuals sampled from a large population, the output of a particular firm, or GDP of a particular country. In this [cross-sectional](#) data setup, we have one observation for each [unit](#)—e.g., person, firm, or country. In [panel data](#) settings we have repeated observations for each unit. For example, we might observe the income of individual i in multiple years or at multiple ages. In these cases, it is common to introduce a second subscript to random variables to make the two dimensions of the data clear: Y_{it} for unit i and time t . Note that as above, i need not denote people. It is perfectly fine to have firms, countries (or maybe even planets) as the units under study. Nor need t denote time specially. The key is simply that for every i there are multiple observations, each of which is indexed by t .

What makes panel data settings different than the cross sectional cases we've discussed so far is that we typically do not assume that Y_{it} and $Y_{it'}$ are independently distributed (e.g., due to persistent, unit-level differences). Likewise, it is common to assume that Y_{it} and $Y_{i't}$ may also not be fully independent (e.g., due to aggregate “time effects”). If everything were independent, then we would be back to a simple cross-sectional case where we could treat each it combination as a distinct unit. Despite these complications, panel data models are widely used in applied economics because a) many data sets of a panel structure, and b) they allow for different identification strategies that may be more credible than what is possible with cross-sectional data alone. One such strategy that we'll study in this note, difference-in-differences, is probably the most popular strategy used in applied social sciences today.

Panel models

In the cross-sectional case, identification of causal effects via regression requires a conditional independence assumption. That is, it must be that $Y_i(1), Y_i(0) \perp\!\!\!\perp D_i$ (or at least conditionally independent), where D_i is the (in this case) binary treatment. Assuming conditional independence is often unpalatable when we have reason to believe that the units who take up treatment may differ systematically from those who do not. Students who get into Chicago, for example, are not randomly selected, so comparing mean test scores of Chicago grads vs. Northwestern grads is unlikely to be informative about the causal effects of attending Chicago on test scores.

With panel data, it is sometimes possible to weaken these assumptions by control for all *time-invariant* factors that differ between units and might confound the treatment. Typically this is accomplished by including “fixed effects” (i.e. indicators $1\{i = 1\}, \dots, 1\{i = N\}$), which dummy out each of the units, in the control vector X_i . With these controls, we would only have to worry about time-varying unobservables which might induce correlations between potential outcomes and treatment.

To be specific, let $Y_{it}(1)$ and $Y_{it}(0)$ denote the potential outcomes of unit i at time t corresponding to their treatment status at time t , D_{it} . Observed outcomes are given by:

$$Y_{it} = (1 - D_{it})Y_{it}(0) + D_{it}Y_{it}(1) = Y_{it}(0) + \beta D_{it}$$

where I have imposed constant treatment effects for simplicity. Suppose further that $Y_{it}(0) = \alpha_i + \epsilon_{it}$, where α_i is a parameter that reflects the unit specific mean of $Y_{it}(0)$ over t , so that $E[\epsilon_{it}|\alpha_i] = 0 \forall i$. Suppose also that $Cov(D_{it}, \epsilon_{it}) = 0$, so that treatment choices are uncorrelated with deviations from this unit specific mean. For example, in the Chicago vs. Northwestern example, the Chicago admits may be higher ability on average but the choice to come to Chicago cannot be related to deviations from this average—whether you’re having a good or bad year at time t , for example. Then the model is:

$$Y_{it} = \beta D_{it} + \alpha_i + \epsilon_{it}$$

A few things to note here. First, there are a lot more parameters than we are used to seeing. Not only do we have the parameter of interest β , but we have one α term for every unit i . That is potentially a lot of parameters! Second, there is no constant term here. In fact, we don’t need one, since we have an α_i for every i , so a constant term would be perfectly colinear (it is the sum of all unit indicators). Finally, notice that is automatically going to be a model where independence across Y_{it} and $Y_{it'}$ will not hold generally. These two random variables are linked by the α_i .

It is also common to assume that ϵ_{it} has a common time component, so that $E[\epsilon_{it}|\alpha_i] = \gamma_t$. Now we have introduced a second dimension of dependence in the model, which can write as:

$$Y_{it} = \beta D_{it} + \alpha_i + \gamma_t + \nu_{it} \tag{1}$$

where $\nu_{it} = \epsilon_{it} - \gamma_t$. The γ_t terms are another potentially large set of parameters (one for each t) that allow units to experience common “shocks”. In a year where the test makers cook up a particularly difficult exam, for example, all students at both Northwestern and Chicago may do worse. Allowing for these time shocks further weakens our identifying assumptions by making sure we are not confounded by correlation between treatment take up and time-specific variation in potential outcomes, in the same way that the α_i ensure we are not confounded by correlation between treatment take up and persistent differences across units.

As written, however, it is not possible to estimate any of the parameters of this model (β , the α_i s, or the γ_t s). Since the α_i s and the γ_t s are unknown parameters, there is no way to “include” them as additional controls. Next we will discuss the two most common strategies for circumventing this challenge.

Fixed effects

Notice that our baseline panel model in Equation 3 can also be written as:

$$Y_{it} = \beta D_{it} + \sum_{i'} \alpha_{i'} 1\{i = i'\} + \sum_{t'} \gamma_{t'} 1\{t = t'\} + \nu_{it} \quad (2)$$

That is, α_i and γ_t can be thought of as the coefficient on a set of indicator functions that take the value of 1 when observation belongs to the appropriate i or t respectively. These “fixed effects” can be constructed as regressors. One can then estimate the sample analogue of Equation 2 by regressing Y_{it} on D_{it} a set of indicators for each unit i and a set of indicators for each time period t .

To see why this might work, consider what would happen if we first tried to “partial out” the α_i by regressing the other covariates on the set of $1\{i = i'\}$ fixed effects. The fitted values from regressing anything on a set of indicators for each unit will be the unit-specific means (**exercise: prove / check this**). So after applying FWL, our model will be:

$$\begin{aligned} \tilde{Y}_{it} &= \beta \tilde{D}_{it} + \sum_{t'} \tilde{\gamma}_{t'} 1\{t = t'\} + \tilde{\nu}_{it} \\ (Y_{it} - \bar{Y}_i) &= \beta (D_{it} - \bar{D}_i) + \sum_{t'} \gamma_{t'} (1\{t = t'\} - \bar{t}_i) + (\nu_{it} - \bar{\nu}_i) \end{aligned}$$

It is common to be dealing with **balanced** panel data, meaning that each unit i has exactly one observation in each of $t \in \{1, \dots, T\}$ periods. In this case, the unit-level average of $1\{t = t'\}$ is simply $1/T$ for every i , so that demeaning effectively simply takes out a constant. Intuitively, because each unit has exactly one observation for each t , $1\{t = t'\}$ is uncorrelated with $1\{i = i'\}$.

The effect of demeaning Y_{it} and D_{it} , however, is different, even in a balanced panel data setting. It subtracts the person-level average from both, transforming both the outcome and key regressor to represent deviations from these person-specific means. The regression then asks how deviations from average outcomes *within* each person are correlated with deviations from average treatment. You can sense (I hope) how this strategy might eliminate potential confounds due to comparisons across people with different α_i , since now we’re making comparisons within each i only.

The $\tilde{\nu}_{it} = \nu_{it} - \bar{\nu}_i$ term might confuse you at first. I’ve included this because Equation 2 is the causal model (just another way of writing Equation 3), but we’re imagining applying FWL in a dataset with $T < \infty$. Taking averages over the finite T , $\bar{Y}_i = \beta \bar{D}_i + \alpha_i + \sum_{t'} \gamma_{t'} \bar{t}'_i + \bar{\nu}_i$, with $\bar{\nu}_i \neq 0$ because our assumptions only imply $E[\nu_{it} | \alpha_i, \gamma_t, D_{it}] = 0$. In essence $\bar{Y}_i$ is an *estimate* of the population average of Y_{it} for individual i , with sampling error $\bar{\nu}_i$.

Because the regressor in this case is effectively $(D_{it} - \bar{D}_i)$, consistent OLS estimation of β requires that $Cov(D_{it} - \bar{D}_i, \nu_{it} - \bar{\nu}_i) = 0$. Since $\bar{D}_i = \frac{1}{T} \sum_{t=1}^T D_{it}$ and $\bar{\nu}_i = \frac{1}{T} \sum_{t=1}^T \nu_{it}$, both $\bar{D}_i$ and $\bar{\nu}_i$ include *every* observation of D_{it} and ν_{it} . For this strategy to work, we therefore also need a stronger assumption than simply $Cov(\nu_{it}, D_{it}) = 0$, which is what we had assumed above. Specifically, we need **strict exogeneity**, which requires that:

$$E[\nu_{is} | \alpha_i, \gamma_t, D_{it}] = 0 \quad \forall s$$

This assumption says that *all* realizations of the structural residuals are independent of D_{it} . Strict exogeneity is not an innocuous assumption. It supposes that the process generating the structural errors is entirely unrelated to the process generating the regressors. An good example comes from Mundlak (1961), who considers a model for farmers' production functions:

$$Q_{it} = m_i + \gamma_t + X'_{it}\beta + u_{it}$$

where Q_{it} is the farm's output in a year, m_i is the unobserved quality of the farm manager, γ_t is a series of aggregate shocks such as the weather, X_{it} is a vector of farm inputs (labor, capital, fertilizer, etc.), and u_{it} is a set of idiosyncratic shocks to local productivity. Strict exogeneity rules out the possibility that when confronted with several positive shocks to u_{it} the manager will hire more variable inputs X_{it} . This seems unreasonable given that virtually any dynamic model will imply that several positive shocks to u_{it} will lead to the expectation of higher future productivity levels, which will in turn lead the farmer to demand more variable inputs.

One interesting point about these models is that although under strict exogeneity it is possible to estimate β consistently, unless $T \rightarrow \infty$ it is generally not possible to consistently estimate the α_i consistently. In simple case where $Y_{it} = \alpha_i + e_{it}$, α_i is just the individual-level mean. If we only observe $T = 3$ periods for each unit, then treating the individual-level mean as a consistent estimator of α_i would be clearly inappropriate. Adding more i doesn't help, since that just adds more parameters to estimate. The fact that β can still be consistently estimated is a happy accident attributable to the demeaning argument we just went through. This was only possible to do (so simply) because the model was linear, however. In more complicated models the presence of **incidental parameters** (i.e., the α_i) can make consistent estimation of β difficult or impossible.

Some practical notes. First, it is not generally possible to include a full set of indicators for both every unit and every time period. Because every unit belongs to at least one time period, one of the γ_t s will be a linear combination of all of the α_i s. To see why, suppose that we simply consider $Y_{it} = \alpha_i + \gamma_t + e_{it}$, with $E[e_{it}|\alpha_i, \gamma_t] = 0$. We need $\gamma_t = 0$ for some t to pin down the parameters, since otherwise I am free to add δ to every α_i and subtract δ from every γ_t . Stata (and other software packages) will often drop one of the fixed effects for you, but it's better to do it yourself so you can be aware of what's going on. If you don't tell the package to not include a constant term, it will also drop an additional term. As you saw on your last homework, which indicator you drop matters for the interpretation of the coefficients on the included indicators, so be wary.

Second, including a large set of fixed effects as regressors can be computationally cumbersome because inverting the design matrix is tricky when it is a) big and b) sparse, as would be the case with many units in the data. The demeaning argument we made above applies both conceptually and practically. Sometimes it is easier to manually demean all the regressors and the outcome yourself and then run OLS on the demeaned data, since in a lot of software taking out group means is straightforward.

First differencing

It is possible to weaken the strict exogeneity assumption a bit by taking a first differences transformation instead of the within (i.e., demeaning) transformation. Specifically, our causal model

can also be written as:

$$\Delta Y_{it} = \beta \Delta D_{it} + \Delta \gamma_t + \Delta \nu_{it} \quad (3)$$

where $\Delta Z_{it} = Z_{it} - Z_{it-1}$, i.e., the within-unit change in Z from $t - 1$ to t . To consistently estimate β using OLS, we require simply that $Cov(\Delta D_{it}, \Delta \nu_{it})$, which implies that D_{it} need only be independent of ν_{it+1}, ν_{it} , and ν_{it-1} (i.e. a lead and a lag) instead of all ν_{it} . With two time periods, of course, fixed effects and first differencing are the same and therefore require the same assumptions. But with many time periods, they may be different.

It is not too easy to think of settings strict exogeneity fails but first differencing works. One would need timing assumptions that imply the agents (e.g., the farmers in our earlier example) require several periods in order to respond to shocks. Nevertheless, there are cases where the first differences approach could be consistent while the fixed effects approach fails, and examining robustness to estimates of β under both strategies may be a revealing empirical exercise.

Random effects

The perspective that we just developed thinks of α_i as parameters to be estimated or differenced off. An alternative perspective imagines that the α_i are [random effects](#)—essentially part of the structural residual that is correlated within unit (or over time). These models are often combined with a parametric assumption about the distribution of the random effects, resulting in something like:

$$\begin{aligned} Y_{it} &= \beta D_{it} + \alpha_i + e_{it} \\ E[e_{it} | \alpha_i, D_{it}] &= 0 \\ \alpha_i | D_{it} &\sim N(\mu, \sigma) \end{aligned}$$

Because we are assuming that α_i is normally distributed given D_{it} with mean μ and variance σ^2 , it obviously cannot be correlated with D_{it} , so there is no concern about differences in α_i confounding estimates of β . Instead, a random effect model is a model for the variance of the combined residual $u_{it} = \alpha_i + e_{it}$, which is assumed to be uncorrelated with D_{it} , so it is sometimes referred to as a [variance component](#) model. It is often possible to estimate the parameters of the distribution of α_i (e.g., its variance). Doing so can be helpful for estimation and inference in panel data settings and for some prediction exercises that we might have time to get to later in the class.

It is not helpful, however, for identification of β . In fact, treating the α_i as random motivates not including fixed effects at all. So if the true model looks something like Equation 2 and you decide to use a random-effects estimator, your estimate of β may be inconsistent unless D_{it} and α_i are independent. But in that case, you might as well omit the fixed effects, since there's no between-unit confound to be worried about. Some textbooks discuss the use of Hausman tests to distinguish between fixed and random effect models, but we won't dwell on that much here. Better to state clearly how you think about persistent unit-level differences and the correlation with the regressors of interest from the outset.

Difference-in-differences

Difference-in-differences, or “diff-in-diff,” is an application of panel data identification arguments popularized in economics by David Card and Alan Krueger’s famous study of the effects of the minimum wage. Card and Krueger (1994) were interested in the effect of raising the minimum wage on low-wage employment levels, which was as topical then as it is now. To answer this question, they used panel data and a particular policy event: the 1992 statewide increase in the minimum wage of New Jersey, from \$4.25 to \$5.05 (a huge increase, in both relative and absolute terms at the time). They were interested in how this change over time in the minimum wage related to the change in fast food restaurant employment in New Jersey, but they didn’t stop there. They instead combined this change over time with the analogous change in employment in the neighboring state of Pennsylvania, which kept its wage at \$4.25 over the same period. Their analysis thus comprised a comparison of employment between two states (a “difference”) as it evolved over time (“in differences”), before and after the New Jersey increase. Intuitively, they argued, if we were to see a fall in employment following the minimum wage increase in New Jersey which didn’t appear in the Pennsylvania trends, we might causally attribute this change to the minimum wage itself. The Pennsylvania trends, in other words, offered a control group for the New Jersey trends.

To formalize this idea, think about the simple panel model we developed above with two units (e.g., New Jersey and Pennsylvania) and two time periods (e.g., 1992 and 1993). In period one, neither unit are treated. In the second period, unit two is treated. Then the change in outcomes for unit one from period one to period two is given by:

$$Y_{12} - Y_{11} = \alpha_1 + \gamma_2 - \alpha_1 - \gamma_1 = \gamma_2 - \gamma_1$$

The change in outcomes for unit two over the same periods is given by:

$$Y_{22} - Y_{21} = \beta + \alpha_2 + \gamma_2 - \alpha_2 - \gamma_1 = \beta + \gamma_2 - \gamma_1$$

It is clear then that if we take the differences in these two differences, we will be left with:

$$(Y_{22} - Y_{21}) - (Y_{12} - Y_{11}) = \beta$$

In the Card and Krueger case, if employment outcomes for restaurants in New Jersey and Pennsylvania followed the model in Equation 3, we could therefore estimate the effect of the 1992 change by taking change in average outcomes from before vs. after the change in NJ and subtracting the same difference in Pennsylvania. Since these are all differences in means, with data on employment it is straightforward to construct a standard error and do inference on the estimated treatment effect.

In practice, diff-in-diff is often done with groups of treated and non-treated units. The estimated effect is the pre- vs. post change in average outcomes among the treated units vs. the pre- vs. post change for the controls. Estimating this differences can be accomplished by running the following regression:

$$Y_{it} = \alpha + \omega D_i + \gamma Post_t + \beta D_i \cdot Post_t + e_{it} \tag{4}$$

I leave it as an **exercise** to show that this saturated regression will fit all four relevant means and that β corresponds to the desired diff-in-diff. Though we have written a regression, the idea is the same. The first, within-group differences takes care of any persistent unit-level differences (i.e., α_i s). The change in outcomes from pre to post for the untreated units serves to adjust for time trends (i.e., the change in γ s) among the treated units.

As usual, for identification of β in this model via OLS, we require that the regressors are uncorrelated with e_{it} . To make this clear, consider a setting with two periods again, where no units are treated in period one, and where some units are treated in period two. Then taking first differences of Equation 4 implies:

$$\Delta Y_{it} = \gamma + \beta D_i + \Delta e_{it}$$

We therefore require that $Cov(D_i, \Delta e_{it}) = 0$. This assumption requires that treatment take-up is uncorrelated with *changes* in the residual. So it cannot be that units taking up treatment are on different trends or that, for example, New Jersey adopts a new minimum wage law *because* of changes in restaurant employment.

It is very much in vogue to consider differences-in-differences in models with potentially heterogeneous treatment effects. Using slightly different notation also helps clarify when diff-in-diff might actually be good idea. Specifically, letting $Y_{i,pre}$ $Y_{i,post}$ be the outcomes of unit i pre and post treatment, respectively, we can write the basic diff-in-diff estimand as:

$$\begin{aligned}\beta_{DiD} &= E[Y_{i,post} - Y_{i,pre} | D_i = 1] - E[Y_{i,post} - Y_{i,pre} | D_i = 0] \\ &= E[Y_{i,post}(1) - Y_{i,pre}(0) | D_i = 1] - E[Y_{i,post}(0) - Y_{i,pre}(0) | D_i = 0]\end{aligned}$$

With a little bit of addition / subtraction, we can re-write β_{DiD} as:

$$\beta_{DiD} = \underbrace{E[Y_{i,post}(1) - Y_{i,post}(0) | D_i = 1]}_{TOT \text{ in post period}} + \underbrace{E[Y_{i,post}(0) - Y_{i,pre}(0) | D_i = 1] - E[Y_{i,post}(0) - Y_{i,pre}(0) | D_i = 0]}_{Potential \text{ bias}}$$

That is, β_{DiD} is equal to the treatment effect on the treated in the post period plus a second term that is the difference in the change in outcomes *under no treatment* between pre and post periods between the treated and untreated units. The infamous **parallel trends** assumption says that this difference is zero, so that β_{DiD} identifies the TOT. In words, parallel trends requires that changes in average outcomes among treated and untreated units would have evolved in the same way in a counterfactual world where the treated units had not, in fact, been treated.

Notice that parallel trends is guaranteed by Equation 3, which was a model of constant effects. But even if effects are potentially heterogeneous, estimating Equation 4 will still recover β_{DiD} since it is simply the difference in means. That is:

$$\begin{aligned}\alpha &= E[Y_{i,pre} | D_i = 0] \\ \omega &= E[Y_{i,pre} | D_i = 1] - E[Y_{i,pre} | D_i = 0] \\ \gamma &= E[Y_{i,post} - Y_{i,pre} | D_i = 0] \\ \beta &= E[Y_{i,post} - Y_{i,pre} | D_i = 1] - E[Y_{i,post} - Y_{i,pre} | D_i = 0]\end{aligned}$$

Because identification in diff-in-diff rests on the assumption of parallel trends in a counterfactual world without treatment, it is common to try to support this assumption by showing that parallel

trends holds *prior* to treatment. When there are multiple time periods in pre and post, this can be accomplished by plotting average outcomes among treated and untreated units before and after treatment. One hopes to say two series that track each other closely (in changes) for a long time up until the moment of treatment, when a sudden break emerges. A nice example comes from Davis (2004)'s study of the impact of a cancer cluster on home prices in Churchill County, NV in 2010:

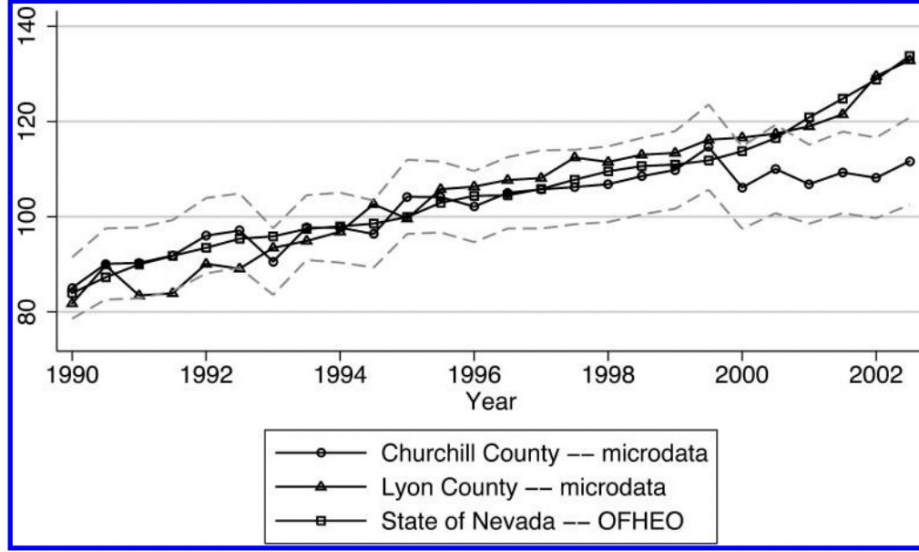


FIGURE 2. ASSESSING THE VALIDITY OF THE COUNTERFACTUAL:
HOUSE-PRICE INDICES FOR SINGLE-FAMILY RESIDENCES, 1990–2002

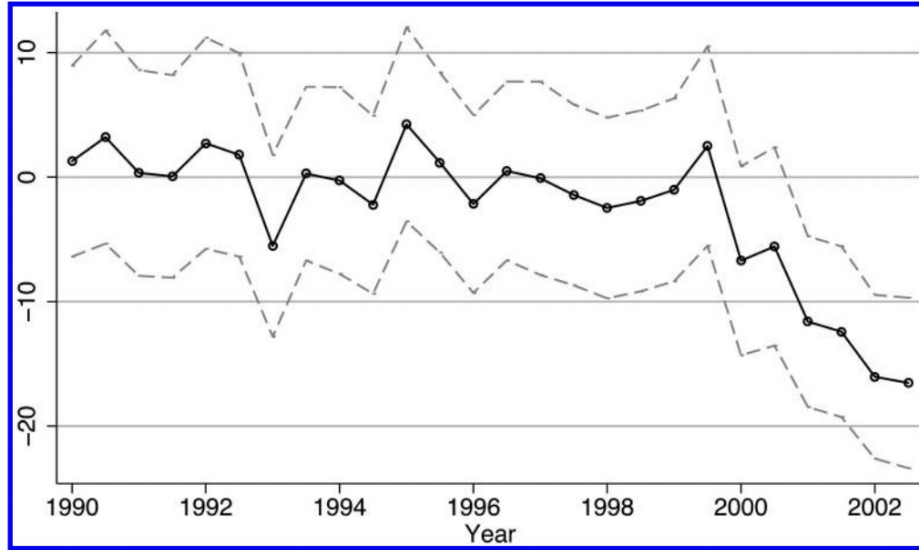


FIGURE 3. FALL IN HOUSING PRICES AFTER 1999:
PERCENTAGE DIFFERENCE BETWEEN CHURCHILL COUNTY HPI AND NEVADA HPI

It is also possible to weaken the parallel trends assumption by including covariates in the model. Note that any time-invariant covariates are not going to play an explicit role since the first difference will account for any level differences across units. That is, the assumption that

$$E[Y_{i,post}(0) - Y_{i,pre}(0)|D_i = 1, X_i] - E[Y_{i,post}(0) - Y_{i,pre}(0)|D_i = 0, X_i]$$

implies that regular (unconditional) parallel trends holds as well. Still, you might want to use

covariates to refine the treatment and control groups in the same sense that one typically trims based on a propensity score: untreated units with similar X s to treated units might have more similar trends to the treated units. This is not a theorem, however, since two units with completely different characteristics can experience common trends in outcomes.

Time-varying covariates will matter, however, so one might assume that only the following version of parallel trends holds

$$E[Y_{i,post}(0) - Y_{i,pre}(0)|D_i = 1, \Delta X_i] - E[Y_{i,post}(0) - Y_{i,pre}(0)|D_i = 0, \Delta X_i]$$

which implies that counterfactual changes would be the same only for treated and untreated units that undergo the same changes in exogenous covariates. One could then estimate a treatment effect by averaging conditional diff-in-diffs estimates, as we did in the cross sectional case. You might be tempted to add covariates to Equation 4 directly as:

$$Y_{it} = \alpha + \omega D_i + \gamma Post_t + \beta D_i \cdot Post_t + \phi X_{it} + u_{it} \quad (5)$$

In this case, β will estimate a causal effect when we have that $E[Y_{it}(0)|X_{it}] = \alpha + \omega D_i + \gamma Post_t + \phi X_{it}$, which should look a lot like the arguments we made about conditional independence earlier. In essence, if $Y_{it}(0) = \alpha + \omega D_i + \gamma Post_t + \phi X_{it} + \tilde{Y}_{it}(0)$, we require only parallel trends in $\tilde{Y}_{it}(0)$. Using the covariates accounts for predictable deviations from overall parallel trends.

Standard error errors

I emphasized above that in the panel data setting, the data should not be treated as iid. Even after accounting for individual and time effects as in Equation 3, it is likely that the ν_{it} are at least correlated within units. This lack of independence invalidates the basic strategies for inference on regression coefficients we studied earlier. Specifically, recall that when $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$, $\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}$, where $\mathbf{U}$ is the vector of regression residuals. A general expression for the variance of $\hat{\beta} - \beta$ is given by $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Sigma\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$, where Σ is the N by N variance covariance matrix of $\mathbf{U}$ and $\mathbf{X}'\Sigma\mathbf{X}$ is the variance of $\mathbf{X}'\mathbf{U}$.

We have already discussed special cases, including homoskedastic settings where $\Sigma = \sigma^2 \mathbf{I}_N$ so that $\mathbf{X}'\Sigma\mathbf{X} = \sigma^2 \mathbf{X}'\mathbf{X}$. In a heteroskedastic but still iid case, Σ remains diagonal, and we have shown how to construct a consistent estimator with $\mathbf{X}'\mathbf{D}\mathbf{X}$, where $\mathbf{D} = \text{diag}(\hat{U}_1^2, \dots, \hat{U}_N^2)$.

In a panel setting, Σ will not be diagonal if there is dependence between some observations. Specifically we have $\mathbf{X}'\mathbf{U} = \sum_{i=1}^N \sum_{t=1}^T X_{it} U_{it}$. If we assume that the data are iid across i but not t , we can think of a central limit theorem (with appropriate $\sqrt{N}$ normalization) applying to the sum $\sum_{i=1}^N \overline{XU}_i$, where $\overline{XU}_i = \sum_{t=1}^T X_{it} U_{it}$. These are iid because we have assumed that dependence only arises across t *within* each i . In essence, you can imagine that we sampled each i iid from a super-population, and each i came with a set of T potentially correlated observations.

By independence, the variance of $\mathbf{X}'\mathbf{U} = \sum_{i=1}^N \overline{XU}_i = \sum_{i=1}^N \text{Var}(\overline{XU}_i)$. The variance of $\overline{XU}_i$ is the variance of $\sum_{t=1}^T X_{it} U_{it}$. Recall that X_{it} is a k by 1 vector, so this variance will be a k by k matrix. We can write this variance matrix compactly in matrix notation as $X_i' \Sigma_i X_i$, where X_i is the T by K matrix of T observations for unit i and Σ_i is the potentially non-diagonal T by T variance covariance matrix of the U_{it} s for unit i .

Without strong assumptions, we can't estimate Σ_i for every i consistently since we only have a finite number of observations of each (T , to be precise). Notice, however, that if we order $\mathbf{U}$ by units i , due to our assumptions $Var(\mathbf{X}'\mathbf{U}) = \mathbf{X}'\Sigma\mathbf{X}$, where

$$\Sigma = \begin{pmatrix} \Sigma_1 & 0 & \dots & 0 \\ 0 & \Sigma_2 & \dots & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & \dots & \Sigma_N \end{pmatrix}$$

has this block diagonal structure. Then $\mathbf{X}'\Sigma\mathbf{X} = \sum_{i=1}^N X_i'\Sigma_i X_i$, where X_i is the vector of T observations for unit i . One can show that in this case:

$$\frac{1}{NT} \sum_{i=1}^N X_i' \hat{U}_i \hat{U}_i' X_i \xrightarrow{p} \lim_{N \rightarrow \infty} \frac{1}{NT} \sum_{i=1}^N E[X_i' \Sigma_i X_i]$$

where $\hat{U}_i$ is the T by 1 vector of $\hat{U}_{it}$ for unit i .

We can then construct a “cluster-robust” estimator of the variance-covariance matrix of $\hat{\beta}$ using this idea as:

$$\hat{V}_{clust} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^N X_i' \hat{U}_i \hat{U}_i' X_i \right) (\mathbf{X}'\mathbf{X})^{-1}$$

Notice that consistency here requires the number of units (the i s) to get large. So in a panel setting, we need to add more people to the data, not more time periods, when clustering by person. Our estimate of $\hat{V}_{clust}$ is likely to be bad when N is small (even when T is big), because our estimate of the “meat” of this sandwich formula only converges to its probability limit when N is big.

In practice, you won't have to compute this object manually. Most statistical software allows for a clustering option. In Stata, for example, one would use `cluster(clustervar)` as an option. Notice also that clustered standard errors are also heteroskedastic robust—we don't impose that Σ_i has common variance components on the diagonal, for example. Also notice that when $T = 1$, so that we are clustering on each observation, the cluster-robust variance covariance estimator is identical to the classic EHW estimator.

There are other settings besides panel settings where clustering is important for inference. In some cases, you may want to cluster because your sampling scheme is [hierarchical](#). Suppose, for example, that I am trying to learn about average test scores among schools, and I sample N schools from the total population and then observe all test scores for the students in those schools. Here it's appropriate to think about the schools as sampled iid from the super-population and allow student-level residuals to be potentially correlated within school. In other words, you should cluster by school.

In other cases, you may have treatment assignments that are clustered. If I randomly assign an intensive test prep program to students but do so at the school level—so that, for example, all students in each school are either treatment or control—my experimental strategy has effectively induced a bunch of dependence between units in the same school, and clustering would be wise. There is a lot of complexity and nuance here. Discussions about clustering often quickly get

metaphysical. But good practical advice is to always a) cluster you standard errors by unit in a panel setting where you observe the same unit over time, and b) cluster at the level of treatment assignment in experiments.

The clustering issue was made most famous by Bertrand, Duflo and Mullainathan's (2004) QJE article, which showed that 5% hypothesis tests for null effects of a set of fictional "placebo" laws that never actually took place rejected far more than 5% of the time without accounting for clustering. Here's their key table showing rejection rates under a handful of different DGPs, with panel A based on actual CPS data and panel B based on simple simulations. You can see that rejection rates are much, much higher than 5% except when there is no serial correlation in observations.

TABLE II
DD REJECTION RATES FOR PLACEBO LAWS

A. CPS DATA				
Data	$\hat{\rho}_1, \hat{\rho}_2, \hat{\rho}_3$	Modifications	Rejection rate	
			No effect	2% effect
1) CPS micro, log wage			.675 (.027)	.855 (.020)
2) CPS micro, log wage		Cluster at state-year level	.44 (.029)	.74 (.025)
3) CPS agg, log wage	.509, .440, .332		.435 (.029)	.72 (.026)
4) CPS agg, log wage	.509, .440, .332	Sampling w/replacement	.49 (.025)	.663 (.024)
5) CPS agg, log wage	.509, .440, .332	Serially uncorrelated laws	.05 (.011)	.988 (.006)
6) CPS agg, employment	.470, .418, .367		.46 (.025)	.88 (.016)
7) CPS agg, hours worked	.151, .114, .063		.265 (.022)	.280 (.022)
8) CPS agg, changes in log wage	-.046, .032, .002		0	.978 (.007)
B. MONTE CARLO SIMULATIONS WITH SAMPLING FROM AR(1) DISTRIBUTION				
Data	ρ	Modifications	Rejection rate	
			No effect	2% effect
9) AR(1)	.8		.373 (.028)	.725 (.026)
10) AR(1)	0		.053 (.013)	.783 (.024)
11) AR(1)	.2		.123 (.019)	.738 (.025)
12) AR(1)	.4		.19 (.023)	.713 (.026)
13) AR(1)	.6		.333 (.027)	.700 (.026)
14) AR(1)	-.4		.008 (.005)	.7 (.026)